

PHYS40180 Research Project

Mark rubric and grade descriptors

An overview of the mark breakdown can be found on this page. The grade descriptors for each element of the assessment follow on the subsequent pages.

In the case of the Implementation, Thesis and Viva, the grade descriptors for each element are for indication only, since it is impossible to cover all scenarios, given the huge variety of projects on offer. Your primary assessor will be matched as closely as possible to the subject area, and they will interpret the rubric as appropriate to the research area and type of project.

Note that the actual performance can fall between the grade descriptors, and the marks awarded may reflect that some elements described in the grade descriptor were better executed than others.

Only integer marks will be awarded under each criterion in all cases below.

Mathematics and Physics

Assessors will be notified ahead of their assessment when they are marking coursework and interviews relating to Mathematics and Physics students. This information will also be made available to the moderation panel. This is to ensure that the rubric and grade descriptors are correctly interpreted in the context of MP-type projects, and thus appropriately, consistently and fairly applied.

Mark Breakdown

Literature Review Presentation and Q&A	10/100 = 10%
Ethics and Research Integrity Assignment	5/100 = 5%
Implementation (10/100 = 10% of total mark)	
Record keeping	5/100 = 5%
Effort, initiative, critical thinking and originality	5/100 = 5%
Thesis (50/100 = 50% of total mark)	
Structure, presentation and style	5/100 = 5%
Literature review and referencing	5/100 = 5%
Technical content	15/100 = 15%
Interpretation and critical evaluation	15/100 = 15%
Extent and quality of results	10/100 = 10%
Viva (25/100 = 25% of total mark)	
Presentation	5/100 = 5%
Quality of answers	5/100 = 5%
Understanding of the project and results	15/100 = 15%

Grade descriptors

Literature Review Presentation and Q&A	1 Below expectations	2 Satisfactory	3 Above expectations
Quality of presentation	Presentation poorly structured, unclear, difficult to follow, lacking good time management, or supported by visual aids of low quality or relevance.	Presentation well structured, clear, easy to follow and supported by relevant and well prepared visual aids.	Presentation and visual aids at a professional level with little room for improvement.
Extent of literature review	Limited review of the existing literature, or unsuitable choice of sources	Evidence of review of key literature, with suitable sources referenced.	Literature review is noteworthy in its relevance and the critical selection of sources.
Quality of critical analysis	Limited critical analysis, project not motivated or placed in context of the existing literature.	Project motivated and placed within the context of the existing literature.	Critical analysis symptomatic of a detailed and comprehensive oversight of the existing literature.
Discussion of impact	Limited discussion of the potential impact of the project, either academic or non-academic.	Potential impacts of the project identified, either academic or non-academic.	Innovative and holistic assessment of potential impacts provided.

Ethics and Research Integrity Assignment	1 Below expectations	2 Satisfactory	3 Above expectations
Quality of log	Limited record that does not evidence the implemented workflow.	Annotated and with sufficient detail to reconstruct the implemented workflow.	Highly detailed log that provides a coherent record of the workflow and the rationale for the refinements.
Quality of critical appraisal	Limited critical appraisal of the generated theory of change.	Critical appraisal of the output used to inform the refinement or identify weaknesses and strengths of the generated output.	Highly detailed critical appraisal of the output with reference to own thoughts, and external sources and literature.
Quality of reflection	Limited reflection.	Answers evidence reflection, informed by the workflow and outputs.	Coherent and insightful reflection, informed by the workflow and outputs, and external sources and literature.
Quality of presentation	Presentation poorly structured, unclear, difficult to follow, lacking good time management, or supported by visual aids of low quality or relevance.	Presentation well structured, clear, easy to follow and supported by relevant and well prepared visual aids that refer to the implemented workflow, critical appraisal of the output, and reflective practice.	Presentation, visual aids, and coherent well-selected content at a professional level with little room for improvement.

					A mark ~ 6 should be considered average.	Marks of 7 and above should be considered high marks.		
Implementation	Inadequate < 3	Insufficient 3	Sufficient 4	Satisfactory 5	Good 6	Very good 7	Excellent 8	Exceptional >=9
Record keeping 5/100 This mark should be awarded based on all forms of record keeping used by the student, including handwritten notebooks, electronic notebooks, repositories and code bases.	Missing or inadequate record.	Sparse record. Of no use for future reference.	Basic record, but key aspects are missing, e.g., data, analysis of results, key calculations, or references.	Some elements of "Good" and some elements of "Sufficient".	Contemporaneous record, but some details are missing, e.g., dates or figures.	Comprehensive, robust and contemporaneous record. All relevant details included, no loose sheets or uncollated material, and all records dated.	As before, with aims and objectives, and summaries included.	As before and, based on the records alone, all results can be cross-checked, the project can be fully reconstructed, and a complete thesis can be written. At the standard expected of a professional researcher.
Effort, initiative, critical thinking and originality 5/100	Inadequate effort that limited progress. This could include a serious lack of attendance. Inadequate evidence of initiative, critical thinking or originality.	Progress was slower than expected due to lack of effort or understanding, or limited ability to handle the equipment or methods, even considering the time needed to learn how to use software or apparatus. Instructions or project plans were only partially followed.	Instructions or project plans were followed, but there was only limited evidence of initiative or critical thinking to help shape the project. Evidence of student learning.		Instructions or project plans were executed successfully in a timely fashion. Good evidence of initiative, critical thinking and student learning. A new type (for the student) of result (including null results), data, method or experimental setup has been produced.	The student was in full control of the project, gaining proficiency with the equipment, theory or methods. Their initiative and critical thinking helped to shape the project, and results were critically evaluated and used to inform the project planning.	As before, but the student showed originality that shaped the project.	As before, but the student was able to operate independently from the supervisor with confidence. At the standard expected of a professional researcher.

					A mark ~ 6 should be considered average.	Marks of 7 and above should be considered high marks.		
Thesis	Inadequate < 3	Insufficient 3	Sufficient 4	Satisfactory 5	Good 6	Very good 7	Excellent 8	Exceptional ≥9
Structure, presentation and style 5/100 Theses exceeding a total length of 30 A4 pages (incl. figures but excl. the titlepage, abstract, lists of figures, lists of tables, tables of contents, references or appendices) or fail to adhere to the guidance on presentation of theses should not be awarded more than 40% in this category.	Inadequate attempt at organisation. The thesis is very difficult to follow, due to incoherent narrative or frequent serious errors of English. Poor presentation that seriously limits the accessibility of the thesis. Inadequate use of figures or tables.	Key sections are missing, e.g., the introduction or conclusions. Errors of English may be frequent, and the style is not appropriate. The presentation is of insufficient quality. Poor quality, insufficient, limited or excessive use of figures or tables that is detrimental to the thesis.	The structure is adequate, but the thesis is not logically organised. The style is not appropriate. An abstract summarising the thesis is included. There are problems with the presentation that affect the readability. There are a significant number of minor errors of English. Basic use of figures or tables that is limited or excessive in places. Captions may be missing or uninformative, or plots may be missing labels or units.	Some elements of "Good" and some elements of "Sufficient".	Clear structure is followed, but the organisation or balance could be improved. The writing is in an appropriate scientific style, with only a small number of minor errors of English. A coherent abstract is included. There are no problems with the presentation. Use of figures or tables complements the thesis, but the presentation or quality of the figures could be improved, or captions could be more informative.	The thesis is logically organised with a balance of material between sections. The thesis is written in an appropriate scientific style with only a small number of minor errors of English. A concise, well-structured and informative abstract is included. The presentation aids the readability. Figures or tables are well presented, used to good effect and cross-referenced in the main text.	As before, but the structure adds clarity. The thesis is easy to read, free of errors and an example of a well-developed scientific style. The presentation is professional, including typesetting of all equations. Figures or tables are professionally presented, and all captions are concise and informative.	As before, but signposting and cross-referencing are used to good effect. The thesis is highly informative, engaging and a pleasure to read. There is evidence that figures or tables have been designed and selected to benefit the thesis and best support the analysis and conclusions. At the standard expected of a professional researcher.

					A mark ~ 6 should be considered average.	Marks of 7 and above should be considered high marks.		
Thesis	Inadequate < 3	Insufficient 3	Sufficient 4	Satisfactory 5	Good 6	Very good 7	Excellent 8	Exceptional >=9
Literature review and referencing 5/100	Limited or inappropriate references, with little or no literature review.	Key references and aspects of literature review are missing.	Basic use of references and cursory literature review. References may be missing information, incorrectly cited or inconsistently formatted.	Some elements of "Good" and some elements of "Sufficient".	A literature review is used to give context to the thesis, but the bibliography remains limited. Some information may be missing or more citations might be expected throughout the thesis.	A comprehensive literature review is provided that motivates the thesis. Appropriate range of sources is used, citations are complete, and the thesis is well referenced throughout.	As before, but the bibliography contains an extensive range of sources, including primary, secondary and tertiary literature, as well as classic and recent material.	As before, but there is evidence of a broad command of the existing literature and strong awareness of the current state-of-the-art. At the standard expected of a professional researcher.
Technical content 15/100	Inadequate content, or serious errors in the physics (and mathematics in the case of MP projects) or technical content, including any figures and tables, that negate the results or conclusions.	Detrimental errors in understanding of the physics (and mathematics in the case of MP projects) or technical content or its presentation, including through figures and tables. Explanations incorrect or inaccessible. Introductory or conclusory material cannot be understood by a non-expert. There is a lack of clarity across the majority of the thesis.	Basic explanations of the physics (and mathematics in the case of MP projects) and technical content are given, although there are important gaps. There is evidence of basic understanding, but the introductory and conclusory material remains inaccessible. There is terseness, verbosity, repetition, or lack of clarity that are detrimental to the understandability of thesis. Figures and		The physics (and mathematics in the case of MP projects) and technical content are explained clearly and concisely, there is evidence of good understanding, and the introductory and conclusory material is accessible. Figures and tables are correct and contain all necessary information.	Explanations are clear, concise and correct, evidencing solid understanding. Attention is given to the pedagogy and accessibility of explanations throughout the thesis, and the introductory and conclusory material are understandable to a non-expert. Figures and tables are correct, complete and easily interpreted.	As before, but the explanations demonstrate a deep understanding of the physics (and mathematics in the case of MP projects) and technical content. The attention to pedagogy ensures that the entire thesis can be appreciated by a non-expert.	As before, but the exposition of the physics (and mathematics in the case of MP projects) and technical content shows originality and flair. At the standard expected of a professional researcher.

					A mark ~ 6 should be considered average.	Marks of 7 and above should be considered high marks.		
Thesis	Inadequate < 3	Insufficient 3	Sufficient 4	Satisfactory 5	Good 6	Very good 7	Excellent 8	Exceptional >=9
			tables contain errors or omissions.					
Interpretation and critical evaluation 15/100	No critical discussion. No error analysis, where expected.	Limited critical evaluation of the methods, results or analysis. Incomplete or incorrect error analysis, where expected.	Some critical evaluation of the methods, results or analysis. Basic error analysis, where expected.	Some elements of "Good" and some elements of "Sufficient".	Suitable interpretation and critical evaluation of the methods, results or analysis, but discussion is limited in parts. Sound error analysis, where appropriate.	All methods, results or analysis are critically explained and interpreted in the context of the existing literature. Errors are fully evaluated and discussed, where expected.	As before, but internal and external cross-checks are included and described in detail.	As before, but a thorough evaluation of the robustness and limitations of the results, methods or analysis are given. At the standard expected of a professional researcher.
Extent and quality of work 10/100 This should be based on realistic expectations given the circumstances, as indicated by the supervisor in their feedback. Students should not be penalised for lack of progress due to factors outside their control, e.g., faulty equipment, working without a partner, or having to work remotely.	No results or novel synthesis of existing material.	Inadequate results or limited analysis of existing material.	Basic results produced or basic analysis of existing material.		Robust results or novel synthesis of existing material.	Robust and original results or novel synthesis and critical analysis of existing material.	As before, but the results or synthesis have the potential to contribute to the field of research.	As before, but the results are impactful within the context of the existing literature or state-of-the-art. At the standard expected of a professional researcher.

					A mark ~ 6 should be considered average.	Marks of 7 and above should be considered high marks.		
Viva	Inadequate < 3	Insufficient 3	Sufficient 4	Satisfactory 5	Good 6	Very good 7	Excellent 8	Exceptional >=9
Presentation 5/100	The presentation is inadequate and provides, at most, a confused picture of the project and results.	A poorly prepared presentation that makes it difficult to understand aspects of the project or results.	A basic overview of the project is given, but key information or aspects is missing, e.g., the motivation or conclusions. The presentation may be significantly under or over time, or the slides may not support the delivery.	Some elements of "Good" and some elements of "Sufficient".	A complete and understandable overview of the project is given. There may be room for improvement in the structure, the balance of material, or the time keeping. The slides might be cluttered or unclear.	A clear, concise and confident overview of the project is given, covering the motivation, methods, results and conclusions. The slides are clear and uncluttered, and aid the delivery. The presentation is to time.	As before, but the presentation is well balanced with a thoughtful selection of material.	As before, but the presentation is exceptionally creative, informative and engaging. At the standard expected of a professional researcher.
Quality of answers 5/100	Unable to answer questions.	Attempts made to answer some questions.	Attempts made to answer most questions, but answers incorrect or incomplete.		Attempts are made to answer all questions, with only minor errors or imprecision.	Correct answers are given with confidence to all questions.	As before, but answers are correct, confident and comprehensive.	As before, but the quality of answers is at the standard expected of a professional researcher.
Understanding of results and background physics 15/100 The understanding of all students will be assessed individually, even when working in a pair. The marks can differ between project partners when there is a clear difference in performance.	Evidence of serious confusion about large parts of the background physics (or mathematics in the case of MP projects) or the project work.	Evidence of serious confusion about some of the background physics (or mathematics in the case of MP projects) or the project work.	Evidence of a basic understanding of the background physics (and mathematics in the case of MP projects), the project or the results.		Explanations show solid understanding of the background physics (and mathematics in the case of MP projects), the project and the results.	The student can confidently and accurately explain all aspects of the project and background physics (and mathematics in the case of MP projects).	The extent and delivery of explanations are symptomatic of deep understanding of the project and background physics (and mathematics in the case of MP projects).	As before, but the depth of understanding and proficiency extends to the existing literature, state-of-the-art and implications of the results. At the standard expected of a professional researcher.